Quadratic-phase test: settling the 2^16 curve

This study follows up ../quadphase-20261104. That study's single descriptive 2^16 curve (p = 65521, a = 48322, b = 6694, n = 65167) had Q_EC = 4.7249, above all 5 of its null draws (max 4.708), which happens 1 time in 6 by chance. Protocol protocol.json, QUADPHASE-2P16-20261104, was frozen before data (sha256 d7250cca5e90..., commit 82dd7979). quadphase.py is a byte copy of the frozen QUADPHASE script.

Verdict (frozen rule): NULL, with valid controls. Against 40 fresh negation-closed null draws, Q_EC sits at the median: u = 0.50, null mean 4.750, sd 0.229, max 5.581. The earlier exceedance came from a 5-draw null that happened to fall low.

- **Controls.**
  - The EC replay reproduces the recorded Q_EC exactly.
  - The planted control is 4.1× the null maximum, at α_0.
  - The A/A even/odd difference is 1.2%.
- **Consequence.** The quadratic-phase NULL now covers N ≈ $2^{12-2}14$ (9 curves, decisive) and one curve at 2^16. It carries the same scope as before: coherent single-phase leakage of about 0.3 of the set or more, and uncalibrated against weak curve classes.

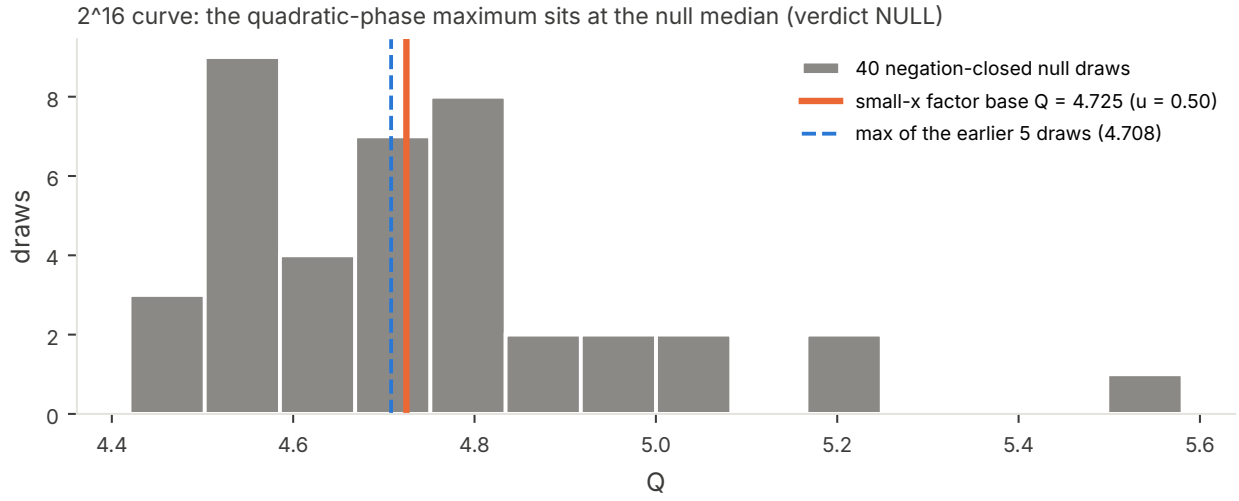

Figure 1: null at 2^16

Deviations

- No S3 checkpoint (no upload token). With the owner's approval, the pushed branch is the resume record.
- No separate independent validation. The study is one curve with a bit-exact replay of a value the QUADPHASE validation had already recomputed independently, and the verdict is a single rank comparison that verdict.json reproduces from rows.jsonl.xz.

Reproduce

python3 rerun_2p16.py --protocol-sha d7250cca5e90d25c8ea0ef17e750bfee09d8c47a28ae6a910fcc9056fccbd1b9 --out rows.jsonl --workers 3 # about 75 min on 3 cores
python3 analyze_2p16.py rows.jsonl verdict.json
python3 figures.py; ./build_pdf.sh